NumPy – Real-World Practice Tasks

Selected Tasks (renumbered 1–3)

Task 1: Student Performance Analysis

You are given the marks of 10 students in 5 subjects.

Columns: Mathematics, Science, English, Computer, Statistics

```
import numpy as np

marks = np.array([
    [85, 78, 92, 88, 76],
    [67, 74, 69, 81, 72],
    [91, 89, 95, 94, 90],
    [45, 52, 48, 61, 55],
    [76, 81, 79, 85, 88],
    [59, 63, 71, 68, 65],
    [88, 92, 84, 90, 86],
    [38, 45, 41, 52, 47],
    [72, 68, 75, 70, 74],
    [95, 91, 89, 96, 93]
])
```

Questions

1. Find the total marks obtained by each student.
2. Find the average marks of each student.
3. Find the highest marks in each subject.
4. Find the lowest marks in each subject.
5. Find the average marks of the class in each subject.
6. Find the student who obtained the highest total marks.
7. Using np.where(), find students whose average marks are greater than or equal to 75.
8. Find students who scored less than 40 in any subject.
9. Create a result array where students with an average of 50 or more are "Pass" and others are "Fail".

Task 2: Employee Salary Analysis

The following array contains the monthly salaries of 8 employees for 3 months.

Columns: June, July, August

```
import numpy as np

salary = np.array([
    [28000, 29500, 31000],
    [35000, 36000, 35500],
    [18000, 19500, 20500],
    [42000, 43000, 44500],
    [25000, 24000, 27000],
    [31000, 32500, 33000],
    [19500, 18500, 21000],
    [38000, 39000, 40500]
])
```

Questions

1. Find the total salary paid to each employee over 3 months.
2. Find the average salary of each employee.
3. Find the total salary paid by the company in each month.
4. Find the month in which the company paid the highest total salary.
5. Find the employee with the highest total salary.
6. Using np.where(), find employees whose average salary is greater than ■30,000.
7. Find all salary values below ■20,000.
8. Increase all salaries below ■20,000 by 10%.
9. Display the updated salary array.
10. Find the overall average salary of all employees.

Task 3: Sales Target & Performance Analysis

A company has 8 sales employees. Their sales for 6 months are given below.

Columns: January, February, March, April, May, June

```
import numpy as np

employee_sales = np.array([
    [85000, 92000, 78000, 105000, 98000, 110000],
    [65000, 72000, 69000, 75000, 81000, 79000],
    [120000, 115000, 128000, 135000, 142000, 138000],
    [55000, 62000, 58000, 67000, 64000, 70000],
    [95000, 88000, 102000, 110000, 108000, 115000],
    [72000, 68000, 75000, 71000, 82000, 85000],
    [45000, 52000, 49000, 58000, 55000, 61000],
    [105000, 112000, 108000, 118000, 125000, 130000]
])

monthly_target = np.array([
    80000,
    85000,
    80000,
    90000,
    95000,
    100000
])
```

Questions

1. Find the total sales of every employee.

2. Find the average monthly sales of every employee.

3. Find the total sales of the company for each month.

4. Find the month with the highest total company sales.

5. Compare each employee's monthly sales with the corresponding monthly target.

6. Using np.where(), create a performance array:

7. "Target Achieved" if sales are greater than or equal to target.

8. "Target Not Achieved" if sales are below target.

9. Find how many months each employee achieved their target.

10. Find employees who achieved their target in at least 4 months.

11. Find the employee with the highest overall sales.

12. Calculate the percentage of target achieved for each employee.

13. Find the employee with the highest average monthly sales.

14. Find the overall average sales of the company.